

Machine learning-assisted ratiometric fluorescence sensor array for recognition of multiple quinolones antibiotics

Mengyuan Li^a, Lei Jia^{a,*}, Xiaolei Zhao^a, Lina Zhang^a, Dan Zhao^a, Jun Xu^{a,*}, Tongqian Zhao^{b,*}

^a College of Chemistry and Chemical Engineering, Henan Polytechnic University, Jiaozuo, Henan 454000, China

^b Institute of Resources & Environment, Henan Polytechnic University, Jiaozuo, Henan 454000, China

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ABSTRACT

Developing analytical methods for simultaneous detection of multiple antibiotic residues is crucial for environmental protection and human health. In this study, a dual lanthanide fluorescence probe (GDP-Eu-Tb) based on nucleotides has been designed. The addition of quinolone antibiotics (QNs) quench the Eu³⁺ fluorescence signal through the inner filter effect (IFE) and exhibit characteristic peaks, enabling ratio fluorescence detection of levofloxacin (LVLX), gatifloxacin (GTLX), and moxifloxacin (MXLX). A ratiometric fluorescence sensor array is constructed using a single sensor element (GDP-Eu-Tb), combined with principal component analysis (PCA) and decision tree (DT) algorithms to model the relationship between fluorescence intensity ratios (I_{450}/I_{616} , I_{460}/I_{616} , I_{463}/I_{616} , I_{468}/I_{616}) and QNs. The performance of the DT model is evaluated using accuracy, precision, recall, and F1 score, with stability and generalizability confirmed by stratified ten-fold cross-validation. This approach demonstrates high sensitivity, selectivity and applicability and provides an effective solution for antibiotic residue detection.

1. Introduction

Since Alexander Fleming's discovery of penicillin and Waxman's initial proposal of the antibiotic definition in 1942 (Chinese academy of sciences, 2011), antibiotics have revolutionized the treatment of human diseases. As one of the major families of antibiotics, quinolone antibiotics (QNs) exhibit broad-spectrum antibacterial properties by inhibiting microbial DNA gyrase and topoisomerase IV, thus blocking microbial DNA replication (Bhatt & Chatterjee, 2022). Due to their significant therapeutic effects, reduced side effects, and cost-effectiveness, they are widely used (Zhou et al., 2022). However, the overuse and improper disposal of QNs have resulted in their active metabolites entering aquatic ecosystems (Lu et al., 2021). These QNs, which are ingested by organisms, are not fully metabolized and tend to accumulate over time. This accumulation contributes to the gradual rise in antibiotic resistance (Zhong et al., 2020), posing significant risks to both ecological systems and human health (Awual, 2015, 2017; Awual et al., 2024). Consequently, monitoring quinolone (QN) residues is crucial for environment protection and safeguarding public health (Liu et al., 2024; Rasee et al., 2023; Salman et al., 2021).

In recent years, various detection technologies have been employed

for antibiotics detection, with widespread applications of methods such as high-performance liquid chromatography (HPLC) (Deng et al., 2022), electrochemical methods (Li et al., 2021), enzyme-linked immunosorbent assays (Ye et al., 2020), surface-enhanced Raman scattering (Shi et al., 2018), and fluorescence methods (Jiang et al., 2024; Xu et al., 2023). However, conventional detection methods, while offering superior sensitivity, are usually complex to operate and require costly instruments, which limits their widespread application in practical scenarios (Awual, 2019; Hossain et al., 2024; Islam et al., 2025; Sheikh et al., 2023). In contrast, fluorescence methods have attracted significant attention due to their simplicity, high sensitivity, and cost-effectiveness (Awual, Hasan, Shahat, et al., 2015; Xu, 2022; Zhang et al., 2021). Nevertheless, most existing fluorescence sensors rely on a specific 'lock-and-key' mechanisms, which limit their effectiveness in detecting multiple antibiotics in complex real-world scenarios (Noreldeen et al., 2022; Yang & Zhang, 2023).

Sensor arrays, also known as 'chemical nose' (Santonocito et al., 2023), enable non-specific detection of multiple target analytes through the interaction between the sensing elements in the array and the analytes (Zhang et al., 2021). By carefully designing and selecting appropriate sensing elements, sensors can simultaneously detect multiple

* Corresponding authors.

E-mail addresses: jljx@hpu.edu.cn (L. Jia), xjil@hpu.edu.cn (J. Xu), zhaotq@hpu.edu.cn (T. Zhao).

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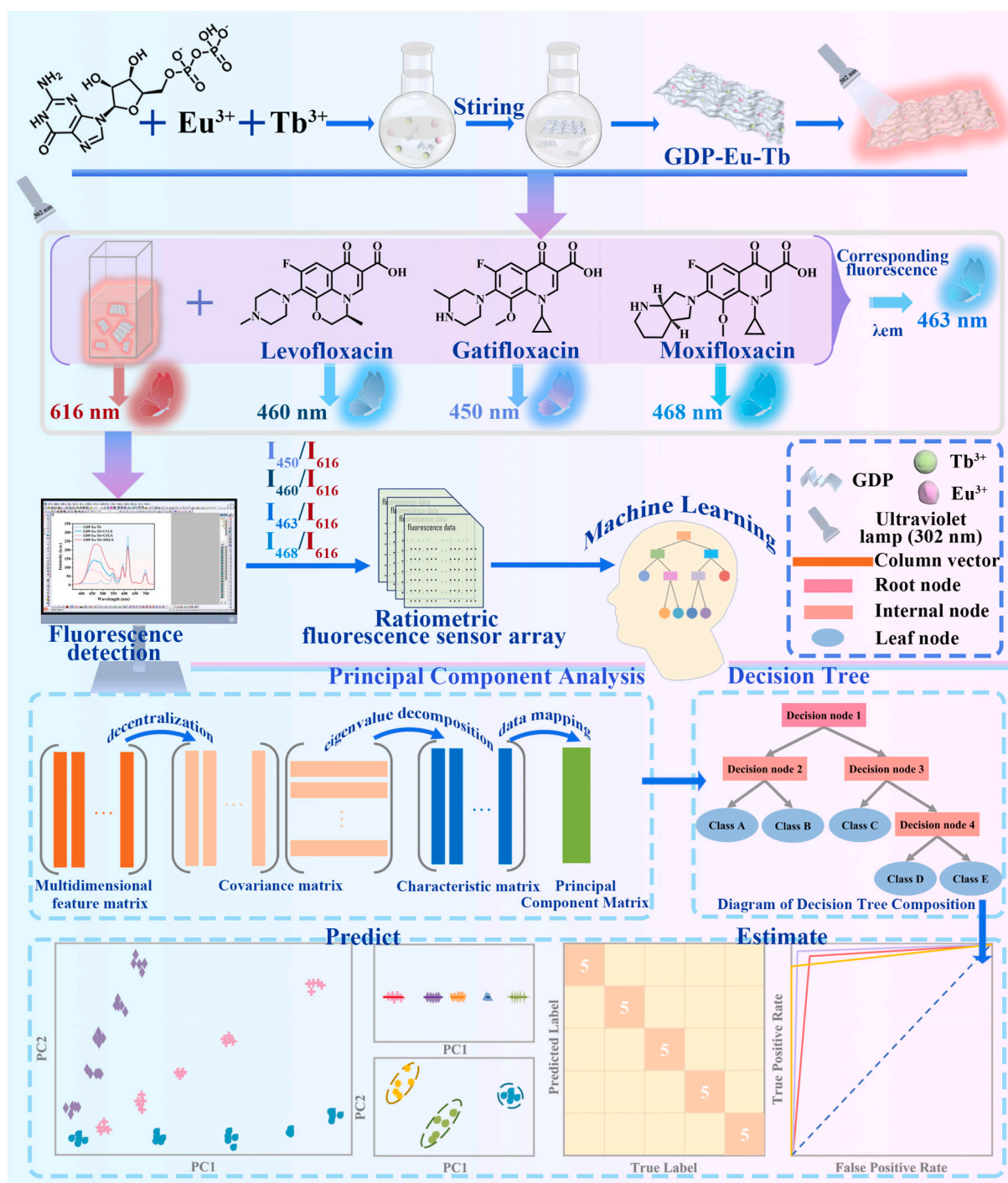
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target analyzers with similar structural characteristics. These sensor arrays have been widely applied in bacterial identification (Santonocito et al., 2023; Sun et al., 2017), meat freshness assessment (Li et al., 2024), and metal ion detection (Zhang et al., 2021). However, existing high-throughput sensor arrays typically consist of multiple sensing elements, which complicates the fabrication process and makes it difficult to efficiently extract large amounts of response data. Moreover, fluorescence sensors that rely on single peak shifts are prone to interference, leading to inaccurate measurements and affecting detection accuracy (Liu et al., 2024; Xu, 2022).

The use of machine learning (ML) algorithms in sensor arrays

enhances data analysis and pattern recognition capabilities, optimizing detection efficiency and predictive accuracy (Lin et al., 2024; Shaban & Alkawareek, 2022). Against this backdrop, this study proposes a ratio-metric fluorescence sensor array with a single multi-signal sensing element, combined with ML algorithms for auxiliary analysis, to achieve the identification and differentiation of various QNs (Scheme 1). The sensor element is constructed through the self-assembly of guanosine diphosphate (GDP) with Eu^{3+} and Tb^{3+} , forming an initial red fluorescence nucleotide-bimetallic lanthanide coordination polymer (GDP-Eu-Tb) for the detection of levofloxacin (LVLX), gatifloxacin (GTLX), and moxifloxacin (MXLX). By calculating the fluorescence intensity ratios of



Scheme 1. A machine learning algorithm assisted a ratiometric fluorescence sensor array based on a double lanthanide GDP-Eu-Tb sensor in analyzing the schematic diagram of QNs.

different QNs relative to the characteristic signal of Eu^{3+} , a pattern for the analysis of various QNs is established, and ML prediction models are constructed using principal component analysis (PCA) and decision tree (DT) algorithms. Finally, the prediction results are visualized and evaluated through scatter plots, confusion matrices, and ROC curves.

Compared to traditional sensors, the development of a ratiometric fluorescence sensor array utilizes a single multi-signal sensing element, combined with ML algorithms, to provide higher sensitivity and accuracy in complex environments (Liu et al., 2024). This approach simplifies the sensor array structure, enhances the efficiency of data analysis, and provides a highly reliable innovative solution for the efficient detection of multiple antibiotics.

2. Experimental section

2.1. Reagents and instruments

Levofloxacin, gatifloxacin, moxifloxacin, europium nitrate hexahydrate ($\text{Eu}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$), Tris (tris (hydroxymethyl) aminomethane), and guanosine-5'-diphosphate disodium salt (GDP) were procured from Shanghai Titan Technology Co., Ltd. Terbium nitrate hexahydrate ($\text{Tb}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) was obtained from Shanghai Aladdin Reagent Co., Ltd. NaOH was acquired from the Hongyan Reagent Factory in Tianjin Hedong District. HCl was sourced from Yantai Shuangshuang Chemical Co., Ltd.

The microstructure of the samples was examined using a Tecnai-G2-F30 transmission electron microscope (TEM) operating at an accelerating voltage of 300 kV, along with energy-dispersive spectroscopy (EDS). X-ray photoelectron spectroscopy (XPS) measurements were conducted using an ESCALAB 250 high-performance electron spectrometer. Powder X-ray diffraction (XRD) data were collected on a Rigaku Dmax 2400 X-ray diffractometer with a scanning speed of $10^\circ/\text{min}$ over a 2θ range of 5° – 80° . Fourier transform infrared spectroscopy (FT-IR) analysis of the sample was performed utilizing a VERTEX 70 infrared spectrometer based on the halide tableting method employing potassium bromide as the reference material. Fluorescence spectrum data were measured using an F97 Pro fluorescence spectrophotometer, while ultraviolet-visible absorption spectroscopy (UV-vis) was carried out with a T2601 UV-visible spectrophotometer.

2.2. Synthesis of GDP-Eu-Tb

Based on the reported synthesis method of GMP-Eu (Wang et al., 2017), an adjustment was made. The GDP solution was prepared by dissolving 4 mg of GDP in 2 mL of Tris-HCl buffer solution (5 mM, pH 9). Subsequently, 5.4 mg of $\text{Eu}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ and 5.4 mg of $\text{Tb}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ were dissolved separately in 1 mL of Tris-HCl buffer solution (5 mM, pH 9), and then added to the GDP solution sequentially, stirring continuously at room temperature 100 min. After centrifugation at $14,800 \times g$ for 3 min, followed by a single wash step, the resulting pellet was dispersed in 2 mL of Tris-HCl buffer solution (5 mM, pH 9) to obtain a super-concentrated stock solution for additional use.

2.3. Fluorescence determination of QNs

Optimizing experimental conditions such as excitation wavelength, pH, volume of the fluorescence probe, and response time are crucial for sensitive detection and accurate identification of LVLX, GTLX, and MXLX. A stable system was formed by combining 80 μL of the ultra-concentrated stock solution of GDP-Eu-Tb with 1920 μL Tris-HCl buffer (5 mM, pH 9). The optimal excitation wavelength was determined by recording the fluorescence spectra of GDP-Eu-Tb before and after adding a mixed solution containing 10 μM QNs using a fluorescence spectrophotometer. Similarly, the most suitable detection pH was confirmed by recording the fluorescence spectra of GDP-Eu-Tb fluorescence probes dispersed in a Tris-HCl buffer solution with pH ranging

from 3 to 12 after adding QNs mixed solution under these optimal conditions. The optimal amount of GDP-Eu-Tb was determined by recording fluorescence spectroscopy data corresponding to different volumes of GDP-Eu-Tb. Finally, under these ideal conditions, the fluorescence spectra of LVLX, GTLX, and MXLX at various concentrations were measured by adding them dropwise into the stable system (2 mL). The concentration of each QN was systematically varied and the corresponding fluorescence intensity was recorded to analyze the detection sensitivity. Images showing the fluorescence of QNs at different concentrations under a 302 nm UV light were captured using the smart-phone camera feature.

2.4. Development of a ratiometric fluorescence sensor array

According to the fluorescence response of GDP-Eu-Tb to QNs, a ratiometric fluorescence sensor array was constructed using only GDP-Eu-Tb for non-specific recognition and differential detection of QNs. Initially, 80 μL of GDP-Eu-Tb stock solution was mixed with 1920 μL Tris-HCl buffer solution (5 mM, pH 9) to establish a stable system. Subsequently, different concentrations of QNs ranging from 0.5 to 30 μM , or a mixture containing 10 μM QNs at varying concentration ratios, were added dropwise and incubated at room temperature for 2 min. The corresponding fluorescence spectra were recorded using a fluorescence spectrophotometer with an excitation wavelength of 290 nm and an emission wavelength range of 360 nm–740 nm (excitation and emission bandwidth: 10 nm; scanning speed: 1000 nm/min).

Due to the strong fluorescence of QNs, the fluorescence spectra of GDP-Eu-Tb exhibited characteristic peaks at 450 nm, 460 nm, and 468 nm upon interaction with different QNs. The consistent fluorescence response of the GDP-Eu-Tb probe towards LVLX, GTLX, and MXLX resulted in a distinct peak at 463 nm, which was used as a response signal. Four groups of fluorescence ratios (I_{450}/I_{616} , I_{460}/I_{616} , I_{463}/I_{616} , and I_{468}/I_{616}) were obtained for QN recognition through ratio logic operation on the response peak intensity. Five replicate experiments were conducted using various concentrations and mixed solutions containing different antibiotics. A ratiometric fluorescence sensor array consisting of four indicators based on intensity ratios $\times$ five replicates $\times$ three antibiotics was constructed. PCA and DT algorithms were then employed to establish an analysis model for QNs using data from this ratiometric fluorescence sensor array, which was subsequently used for analyzing unknown samples. The detection performance of the ML model based on multidimensional datasets acquired from this ratiometric fluorescence sensor array was investigated by identifying different concentrations of single antibiotics, as well as combinations of different antibiotics and QNs.

2.5. Data analysis

The acquired multidimensional data were preprocessed using the Python programming language, and a 60 %:40 % ratio was employed to divide the dataset into training and test sets for model construction and evaluation.

2.5.1. Principal component analysis

The PCA algorithm preserves the most significant information by reducing it to a lower dimension (Li et al., 2023), thereby avoiding issues caused by high dimensionality, facilitating visualization, and is considered an effective data preprocessing technique. In this study, we utilize the PCA algorithm to process the multidimensional dataset and select the optimal number of principal components based on the evaluation results from prediction models for visualizing the analytical results.

2.5.2. Decision tree

The DT library function is based on the sklearn ML library and incorporates several commonly used adjustment parameters. However, in

this study, the constructed DT model only focused on tuning two key parameters: the criterion and max_depth. The optimal values of these selected parameters were obtained by grid search and fine-tuning to optimize the model performance while preventing overfitting. As a result, a more robust DT model with enhanced predictive power was achieved.

2.5.3. Method of evaluation and evaluation metrics

The dataset was stratified into 10 folds using statistical methods to ensure balanced class proportions in each fold, reducing the risk of model bias during training. The optimal number of principal components was determined based on the cross-validation accuracy to enhance the model performance. Various indicators, including a confusion matrix (True label vs Predicted label), receiver operating characteristic (ROC) curve, and area under ROC curve (AUC), were employed to assess the analytical capability of the DT classification model derived from the ratiometric fluorescence sensor array. Furthermore, we evaluated the predictive performance of DT models using multidimensional data from sensor arrays in an interference environment by evaluating classification metrics such as accuracy, precision, recall, and F1 score across different datasets with the same parameters. The equations were calculated as follows.

$$\text{Accuracy} = (TP + TN) / (TP + FP + FN + TN) \quad (1)$$

$$\text{Precision} = TP / (TP + FP) \quad (2)$$

$$\text{Recall} = TP / (TP + FN) \quad (3)$$

$$F1 - \text{score} = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (4)$$

where TP represents the number of correctly predicted positive cases, FP represents the number of incorrectly predicted positive cases, and FN represents the number of incorrectly predicted negative cases.

2.6. Anti-interference test in the presence of cations and anions

In order to evaluate the stability and reliability of the developed fluorescence probe under ionic interference conditions, interferometric tests were performed in different ionic solutions. First, fluorescence spectra were recorded using a fluorescence spectrophotometer to observe the changes before and after the addition of three QNs (10 μM) in 10 μM ionic solutions. Each ionic solution was tested in a triplet to ensure the reliability of the results. In addition, data from two independent sets of five repeated experiments were used for model construction and prediction to further investigate the analytical performance of decision tree algorithms assisted by sensor arrays under complex conditions.

2.7. Treatment and analysis of actual samples

To explore the potential application of the GDP-Eu-Tb fluorescence probe in practical detection, this study employed a spiking sample method to analyze locally purchased honey samples. Firstly, the honey was diluted 30 times with deionized water and filtered through a 0.22 μm membrane to remove solid impurities. Then, five concentrations (0.5, 1, 5, 15, and 25 μM) were selected for the experiment, with each concentration tested in triplicate. Finally, based on the honey sample detection environment, a dataset from five independent replicate experiments was used to construct the model. To further assess the performance of the machine learning-assisted ratiometric fluorescence sensor array in real sample applications, five independent replicate experimental data, completely distinct from the training set, were used as the unknown dataset for predictive analysis.

3. Results and discussion

3.1. The structure and characterization of GDP-Eu-Tb

GDP-Eu-Tb fluorescence nanoprobe have been successfully synthesized at room temperature by self-assembly of di-lanthanide elements and ligands to achieve eye-level visibility for multi-color fluorescence detection. The microstructure and morphology of the fluorescence probe are analyzed using transmission electron microscopy (TEM). Figs. 1A-B demonstrate that the GDP-Eu-Tb fluorescence probe exhibits a closely interconnected network structure due to the multidentate coordination of the GDP ligand (Zeng et al., 2021). The XRD pattern of GDP-Eu-Tb shows no distinct diffraction peak, indicating that the prepared fluorescence probe possesses an amorphous structure (Fig. S1) (Wang et al., 2016). Correspondingly, energy-dispersive spectroscopy (EDS) analysis of GDP-Eu-Tb (Fig. 1C) confirms the presence of P, Eu, and Tb elements, while X-ray photoelectron spectroscopy (XPS) results (Fig. 1D) reveals characteristic peaks corresponding to Eu^{3+} and Tb^{3+} , thereby confirming the successful combination of these two lanthanide elements with GDP. Fluorescence spectra are recorded for GDP-Eu, GDP-Tb, and GDP-Eu-Tb to investigate whether successful coordination between GDP and both lanthanides occurs. As shown in Fig. 1E, a comparison of the same volumn solutions containing a single element combination with GDP at the excitation wavelength of 290 nm reveals a noticeable difference in fluorescence intensity at 616 nm for GDP-Eu and 545 nm for GDP-Tb. This observation indicates a good sensitivity of GDP towards Tb^{3+} . In contrast, the fluorescence spectra obtained for GDP-Eu-Tb show characteristic peaks corresponding to Eu^{3+} and Tb^{3+} , and the observed fluorescence intensity at 616 nm is higher than that at 545 nm. This not only confirms successful coordination between Eu^{3+} and Tb^{3+} but also demonstrates energy transfer between them (Wang et al., 2016; Xu et al., 2019). The confirmation of the phenomenon is further supported by capturing a fluorescence image using a 302 nm ultraviolet lamp (as illustrated in Fig. 1E). To provide additional evidence for the successful coordination between Eu^{3+} , Tb^{3+} , and GDP, Fourier-transform infrared spectroscopy (FT-IR) is conducted on various materials. As depicted in Fig. 1F, the absorption peaks observe at wavenumbers 1698 cm^{-1} , 1484 cm^{-1} , 1237 cm^{-1} , and 1112 cm^{-1} correspond to specific vibrations related to P-OH stretching, guanine C8-N7 stretching, and the anti-symmetric/symmetric vibrations of phosphate groups (Zeng et al., 2021; Zhu et al., 2022). Since the introduction of Eu^{3+} and Tb^{3+} , a slight shift in the peak position is observed at the new values of 1602 cm^{-1} , 1466 cm^{-1} , 1223 cm^{-1} , and 1075 cm^{-1} , respectively. The comparison of FT-IR data indicates that both the nucleobase portion of GDP and the phosphate group coordinate with Eu^{3+} and Tb^{3+} (Zeng et al., 2021).

3.2. The feasibility of GDP-Eu-Tb for the detection of multiple antibiotics and the detection mechanism

In general, the O—H vibrations of water molecules can quench the fluorescence emitted by lanthanide ions (Gao et al., 2017). The incorporation of GDP bridging ligands, self-assembled with lanthanide elements into coordination polymers enables energy transfer from the guanine base to the lanthanide ion, inducing its characteristic fluorescence emission (Gao et al., 2022; Liang et al., 2020). However, GDP based coordination polymers constructed solely with Tb^{3+} or Eu^{3+} have limitations in terms of susceptibility to interference and difficulty in visually observing fluorescence changes. To overcome this issue, di-lanthanide ions (Eu^{3+} , Tb^{3+}) are introduced into the GDP-Eu-Tb fluorescence probe, resulting in five distinct emission peaks corresponding to transitions from $^5\text{D}_4$ to $^7\text{F}_j$ ($j = 6, 5$) for Tb^{3+} and from $^5\text{D}_0$ to $^7\text{F}_j$ ($j = 1, 2, 4$) for Eu^{3+} , respectively (Fig. S2 A) (Chen et al., 2022; Liu et al., 2020; Qu et al., 2020). On one hand, due to the higher energy level of $^5\text{D}_4$ compared to $^5\text{D}_0$ in Eu^{3+} , GDP sensitization towards Eu^{3+} is more efficient (Li et al., 2022). On the other hand, effective energy transfer occurs from Tb^{3+} to Eu^{3+} (Li et al., 2020; Li et al., 2022; Zhang et al., 2024),

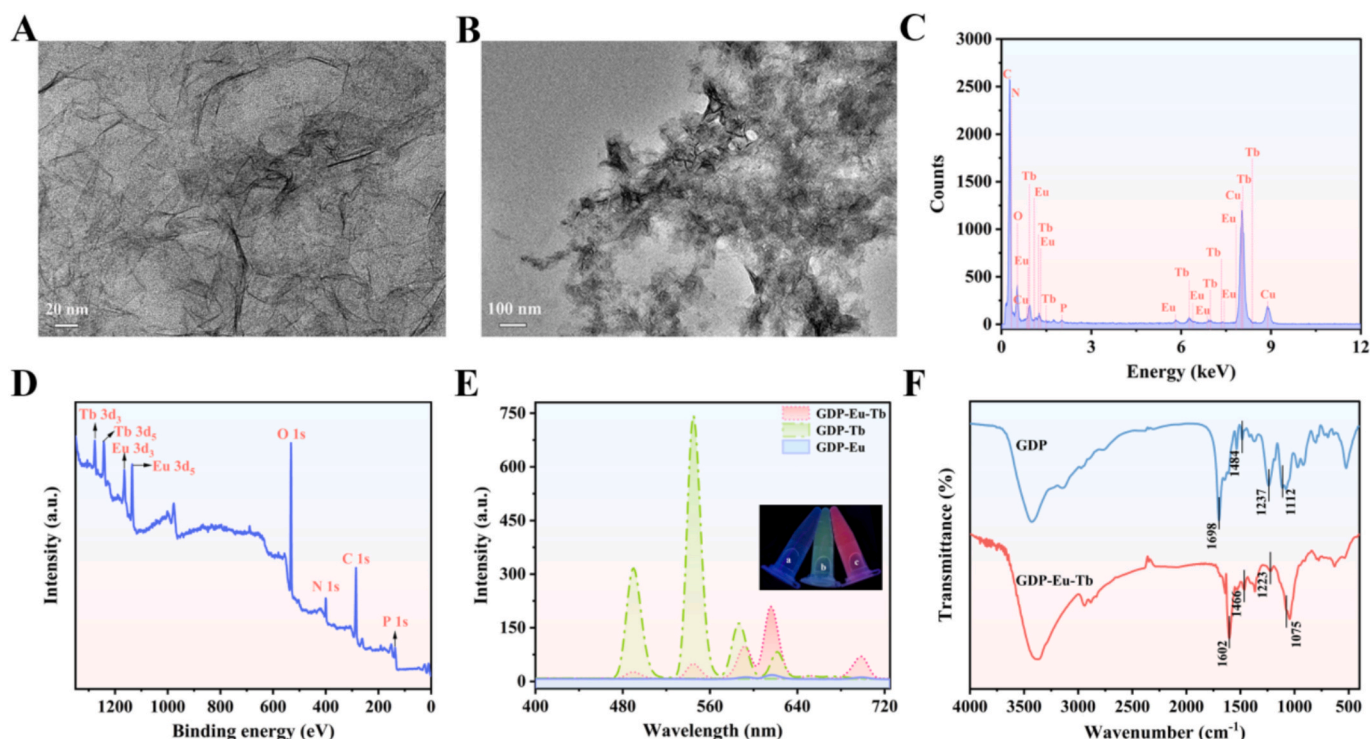


Fig. 1. TEM images of GDP-Eu-Tb at 20 nm (A) and 100 nm (B); EDS energy spectrum (C) and XPS full scan spectrum (D) of GDP-Eu-Tb; fluorescence spectra of GDP—Eu, GDP—Tb, and GDP-Eu-Tb are shown in (E), with the insert displaying fluorescence images of (a) GDP—Eu, (b) GDP—Tb, and (c) GDP-Eu-Tb under a 302 nm ultraviolet lamp; FT-IR spectra of both GDP and GDP-Eu-Tb are presented in (F).

resulting in weaker fluorescence emission from Tb^{3+} . The combined sensitization effect of GDP and Tb^{3+} towards Eu^{3+} make it possible for the GDP-Eu-Tb fluorescence probe to exhibit initial red fluorescence (Fig. S2 A illustration and Fig. S2B), providing a crucial condition for easy visual observation of fluorescence changes during QNs detection.

In order to assess the feasibility of the GDP-Eu-Tb fluorescence probe for detecting various types of QNs, LVLX, GTLX, and MXLX, which have similar structures, are selected as target analytes (Figs. S3A—C). Their effect on the luminescence properties of the synthesized probe is thoroughly investigated. As shown in Fig. 2 A, the addition of 12.5 μM LVLX decreases the fluorescence intensity at 616 nm for GDP-Eu-Tb and introduces a new emission peak at 460 nm. This phenomenon can be attributed to the strong autofluorescence exhibited by LVLX, which manifests as its characteristic peak upon interaction with the GDP-Eu-Tb fluorescence probe (Fig. S4) (Xiao et al., 2021). Similarly, upon adding GTLX and MXLX, distinctive peaks appear at 450 nm and 468 nm, respectively, for the GDP-Eu-Tb fluorescence nanoprobe, while the emission at 616 nm decreases (Fig. 2B and Fig. 2C). Due to these varying degrees of change, different visible fluorescence colors are observed for the GDP-Eu-Tb fluorescence probe under ultraviolet light at a wavelength of 302 nm after exposure to different QNs (illustrated in Figs. 2 A–C). The increase in intrinsic fluorescence intensity of the QNs, coupled with the decrease in fluorescence intensity of the main characteristic peak of Eu^{3+} , enables the quantitative fluorescence detection and analysis of different antibiotics.

The addition of QNs do not result in the emergence of new absorption peaks in the UV absorption spectrum of the detection system (Figs. 2D—F), indicating the absence of a static quenching mechanism between QNs and GDP-Eu-Tb (Aggarwal et al., 2023). However, significant changes are observed in the UV absorption spectra of GDP-Eu-Tb fluorescence probes before and after the addition of QNs, suggesting an interaction between QNs and GDP-Eu-Tb. A fluorescence lifetime decay experiment is performed to calculate the change in the fluorescence lifetime of the 616 nm emission peak of GDP-Eu-Tb upon the

addition of QNs. Data fitting and lifetime calculation are performed using Origin 2021 software with a single-exponential decay model and the average lifetime calculation formula (Eq. (5) and Eq. (6)). Interestingly, it is found that the fluorescence lifetime of GDP-Eu-Tb at 616 nm remained unchanged upon addition of QNs (Figs. 2G–H). This observation suggests that the quenching of the Eu^{3+} fluorescence signal could be attributed to the Inner Filter Effect (IFE). Moreover, as shown in Figs. S5A—C, the absorption peak of the QNs around 290 nm partially overlaps with the excitation spectrum of the GDP-Eu-Tb fluorescence nanoprobe in the 280–320 nm range, but not with its fluorescence emission spectrum. These findings collectively suggest that the IFE induced by QNs leads to a quenching of the fluorescence signal from GDP-Eu-Tb at 616 nm.

$$I = I_0 \exp(-t/\tau) \quad (5)$$

$$\tau_{av} = (A_1 \tau_1^2) / (A_1 \tau_1) \quad (6)$$

here, I_0 is the intensity at time 0, τ_1 is the decay lifetime, and A_1 is the pre-exponential factor of τ_1 time-resolved decay lifetime (Aggarwal et al., 2023).

3.3. Optimization of detection conditions

The above results demonstrate the specific feasibility of the GDP-Eu-Tb fluorescence probe for the detection and analysis of three antibiotics. In order to achieve optimal experimental outcomes and enhance the sensitivity and stability of the sensing signal, several factors are investigated, taking into account the coexistence of these antibiotics. These factors include excitation wavelength, pH of the Tris-HCl buffer solution, probe volume, response time, and fluorescence stability of the fluorescence probe. Fig. S6A shows that when the excitation wavelength is set at 294 nm, the fluorescence intensity at the central characteristic peak of Eu^{3+} (616 nm) reaches its maximum. However, considering that the target antibiotics may cause strong interference with the

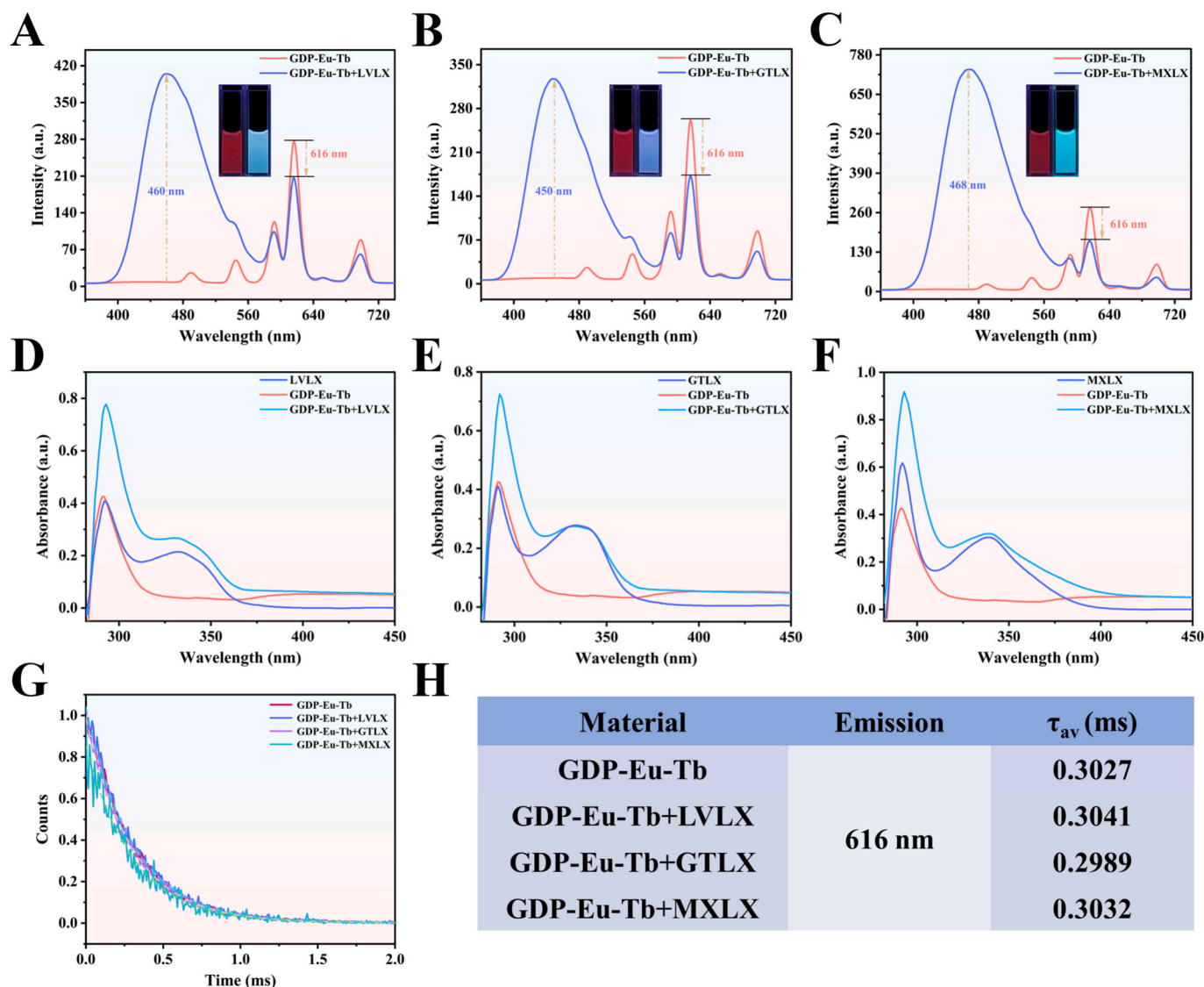


Fig. 2. (A)–(C) The fluorescence spectra of GDP-Eu-Tb before and after the addition of LVLX, GTLX, and MXLX (the insets show the fluorescence images before and after the corresponding antibiotics were added); (D)–(F) UV absorption spectra; (G) the fluorescence lifetime decay curve at 616 nm and the corresponding fluorescence lifetime table (H).

fluorescence signal, further investigation is conducted on the effect of different excitation wavelengths on the fluorescence intensity ratio (I_{463}/I_{616}), as shown in Fig. S6B. The experimental results indicate that the I_{463}/I_{616} fluorescence intensity ratio reaches its maximum at the excitation wavelength of 290 nm, which suggests that this wavelength can effectively suppress interference from the intrinsic fluorescence signal of the antibiotic and optimize the sensitivity of the probe. Therefore, 290 nm is chosen as the optimal excitation wavelength for the detection of the target antibiotic, achieving excellent visual sensing.

In addition, the pH adaptability of the fluorescence probe has a significant impact on its sensing performance (Awual et al., 2017; Awual, Hasan, & Znad, 2015). Fluorescence intensity variations of GDP-Eu-Tb fluorescence probes in a Tris-HCl buffer system are investigated for different values of pH. As depicted in Fig. S7A, the I_{463}/I_{616} ratio rapidly increases within the pH range of 7 to 9, and then significantly decreases beyond this range, indicating that the probe exhibits the best fluorescence response at a pH of 9 (Awual et al., 2016). Therefore, a Tris-HCl buffer with a pH value of 9 is chosen as the optimal detection solvent. Considering that varying probe volumes may influence detection effectiveness, we explore how different probe volumes affect the detection performance for the target antibiotics. The probe volume of

80 μ L corresponds to the maximum value of the I_{463}/I_{616} ratio, as shown in Fig. S7B. Therefore, the optimal probe volume for subsequent detection is determined to be 80 μ L. To investigate the impact of reaction time on the sensing performance of the fluorescence probes, the variation in the fluorescence intensity ratio (I_{463}/I_{616}) over 20 min is recorded. As shown in Fig. S8A, the fluorescence intensity remains constant during this period in the absence of the target antibiotic, indicating good fluorescence stability of the probe. After the addition of the target antibiotic, I_{463}/I_{616} remains stable between 2 and 12 min, demonstrating that the GDP-Eu-Tb fluorescence probe is suitable for on-site detection of the three target antibiotics. Therefore, a reaction time of 2 min is considered optimal.

To further evaluate the long-term fluorescence stability of GDP-Eu-Tb, a comparison is made between the newly synthesized probe and the GDP-Eu-Tb stored at 4 $^{\circ}$ C for 7 months. As shown in Fig. S8B, although the fluorescence intensity of GDP-Eu-Tb decreases by more than 20 % after seven months of storage, its fluorescence signal remains stable over a 20-min period. This indicates that the fluorescence probe can maintain its fluorescence stability for a relatively long period under refrigeration (Hidayat et al., 2017). However, prolonged storage may still lead to some fluorescence attenuation. Therefore, to ensure the

accuracy and reliability of the detection results, it is recommended to avoid storage periods longer than 7 months.

3.4. Optical and visual sensing

Based on the optimal detection conditions outlined above, this study investigates the ratiometric fluorescence response of the GDP-Eu-Tb fluorescence nanoprobe to three target analytes through fluorescence titration. As illustrated in Fig. 3A, with incremental additions of LVLX (0–40 μ M), the intensity of the characteristic peak at 616 nm for Eu³⁺ gradually decreases, while the fluorescence intensity at 460 nm, corresponding to LVLX, significantly increases. Similarly, the GDP-Eu-Tb fluorescence probe shows comparable ratiometric fluorescence responses to GTLX and MXLX. As shown in Fig. 3B and Fig. 3C, with increasing concentrations of GTLX and MXLX, their respective characteristic peak intensities are markedly affected (λ_{exGTLX} : 450 nm; λ_{exMXLX} : 468 nm). Within the concentration range of 0–40 μ M, a strong linear correlation is observed between the added LVLX concentration and the fluorescence intensity ratio (I_{460}/I_{616}), described by the eq. $Y = 0.1550 X + 0.0292$, with a correlation coefficient (R^2) of 0.9946 (Fig. 3D). The p -values, calculated from 14 sample points, are less than 0.001, indicating that the linear relationship between the target analyte concentration and the fluorescence intensity ratio is highly statistically significant. This provides a robust foundation for the subsequent development of sensor arrays and prediction using ML models. The detection limit, based on a three-fold signal-to-noise ratio, is found to be 8.39 nM. As illustrated in Fig. 3E and Fig. 3F, a strong linear relationship is observed between the fluorescence intensity ratio and the corresponding concentrations in the range of 0–40 μ M (GTLX) and 0–30 μ M (MXLX), respectively (GTLX: $Y = 0.1401 X + 0.0349$, $R^2 = 0.9960$, $p < 0.001$; MXLX: $Y = 0.3566 X + 0.0223$, $R^2 = 0.9937$, $p < 0.001$). The detection limits are 9.51 nM (GTLX) and 4.25 nM (MXLX), respectively.

The chromaticity diagram coordinates (Table S1) at different concentrations are calculated using the fluorescence emission spectrum data at an excitation wavelength of 302 nm. These coordinates are drawn using the chromaticity diagram developed by the International Commission on Illumination (CIE). As shown in Fig. 3G, the fluorescence color shifts from the red region to the blue region with increasing LVLX concentration. Under the irradiation of a 302 nm ultraviolet lamp, fluorescence images of GDP-Eu-Tb at different concentrations of LVLX show a transition from red fluorescence to blue fluorescence (Fig. 3J), which is consistent with the coordinate point variation trend in the CIE diagram. The chromaticity diagrams (Fig. 3H and Fig. 3I), derived from chromaticity coordinates (Table S2 and S3) and fluorescence images under ultraviolet light (Fig. 3J) verify the fluorescence color change after the addition of GTLX and MXLX to GDP-Eu-Tb, respectively.

In this study, the developed double lanthanide fluorescence probe demonstrates a broad detection range and a low detection limit for the target QNs. Specifically, the probe enables visual detection and sensitive analysis of QNs by visible light variations from red to blue-green under normal environmental conditions. To evaluate the detection performance of the di-lanthanide probe developed in this study, a comparison is made with detection methods reported by additional researchers. As shown in Table S4, although the Al³⁺/GSH-CuNCs complex (Jiang et al., 2024) has a lower detection limit for GTLX, its detection range is narrower compared to the probe developed in this study. These findings highlight the broader application potential of the GDP-Eu-Tb fluorescence probe.

3.5. Ratiometric fluorescence sensor arrays for recognition of multiple antibiotics

The GDP-Eu-Tb fluorescence probe demonstrates intrinsic fluorescence upon interaction with various antibiotics, while different antibiotics are found to quench the fluorescence signal of Eu³⁺ at 616 nm to varying degrees. Therefore, by exploiting the ratio of the intrinsic

fluorescence signal from the antibiotic to the signal at 616 nm, it is possible to develop arrays of ratiometric fluorescence sensors for the identification of multiple QNs. This sensor array requires only the GDP-Eu-Tb fluorescence probe as its sole sensing element. By employing sensor arrays with four distinct fluorescence intensity ratios across five replicates and varying concentrations or concentration ratios, a comprehensive multivariate data set is generated for subsequent model development. Leveraging ML algorithms enables simultaneous and high-throughput analysis of multiple targets.

The acquisition of information across multiple scenarios often prolongs data collection times and complicates the problem due to the sheer volume of data, leading to challenges associated with the ‘curse of dimensionality.’ PCA is a widely statistical technique for data dimensionality reduction (Lin et al., 2024). PCA works by transforming potentially correlated features from the original dataset into a smaller number of uncorrelated variables, known as principal components (PCs), through linear transformations. These components are typically selected based on their cumulative contribution rate, which represents the amount of variance explained by the components (Lever et al., 2017). However, while PCA effectively reduces dimensionality, it may not be effective for simultaneous identification and predictive analysis of QNs due to the complex nature of multidimensional data obtained from arrays of ratiometric fluorescence sensors. To overcome this limitation, ML algorithms can be employed to model the PCA-preprocessed data, enabling more efficient identification and classification predictions for various antibiotics. As a supervised learning approach, DT algorithms derive rules from datasets containing features and labels by recursively constructing a decision tree that applies a ‘divide and conquer’ strategy to make decisions on unknown data. This methodology is frequently applied to address classification and regression challenges. To enhance the predictive analysis concerning antibiotics, the DT algorithm is implemented in this study to develop a model based on PCA processed data, which is subsequently evaluated for its efficacy in predicting unknown sample outcomes. The DT model construction flowchart is shown in Fig. S9, illustrating the step-by-step process of the model development.

Figs. 3A–C illustrates the titration analysis of the GDP-Eu-Tb fluorescence probe for LVLX, GTLX, and MXLX, which have similar structural characteristics. Based on a specific range of concentration detection, we investigate for the first time the analytical performance of a DT model that utilizes an array of ratiometric fluorescence sensors to identify and quantify these three distinct antibiotics. As feature selection plays a crucial role in model performance, we focus on GTLX as an example to examine how factor selection-derived from PCA preprocessing-affects model construction. During the model construction process, by using stratified ten-fold cross-validation method, the dataset is divided in a way that maintains class distribution in each fold, ensuring better generalization and reducing the risk of overfitting or underfitting (Kucheryavskiy et al., 2020). Table S5 presents the accuracy metrics of the DT model based on various features under stratified ten-fold cross-validation. The results show that the model using the principal component 1 (PC1) with the highest cumulative contribution rate demonstrates a more outstanding predictive ability. A comparison between confusion matrices derived from both training and test sets (Fig. S10A and Fig. S10B) confirms that the DT model demonstrates robust generalization performance and fit. Thus, PC1 is selected as the basis for constructing the DT model. Figs. 4A–C present a one-dimensional classification diagram derived from the DT model constructed using PC1 to differentiate between LVLX, GTLX, and MXLX at varying concentrations. By applying a PCA dimensionality reduction algorithm to pre-process the multidimensional dataset, the DT model effectively predicts various antibiotic concentrations by accurately classifying and predicting the unknown concentration data set. Notably, distinct clusters representing different antibiotic concentrations are obviously visible in the scatter plot (Figs. 4A–C). The intra-cluster distances within each prediction cluster are similar, while the inter-cluster

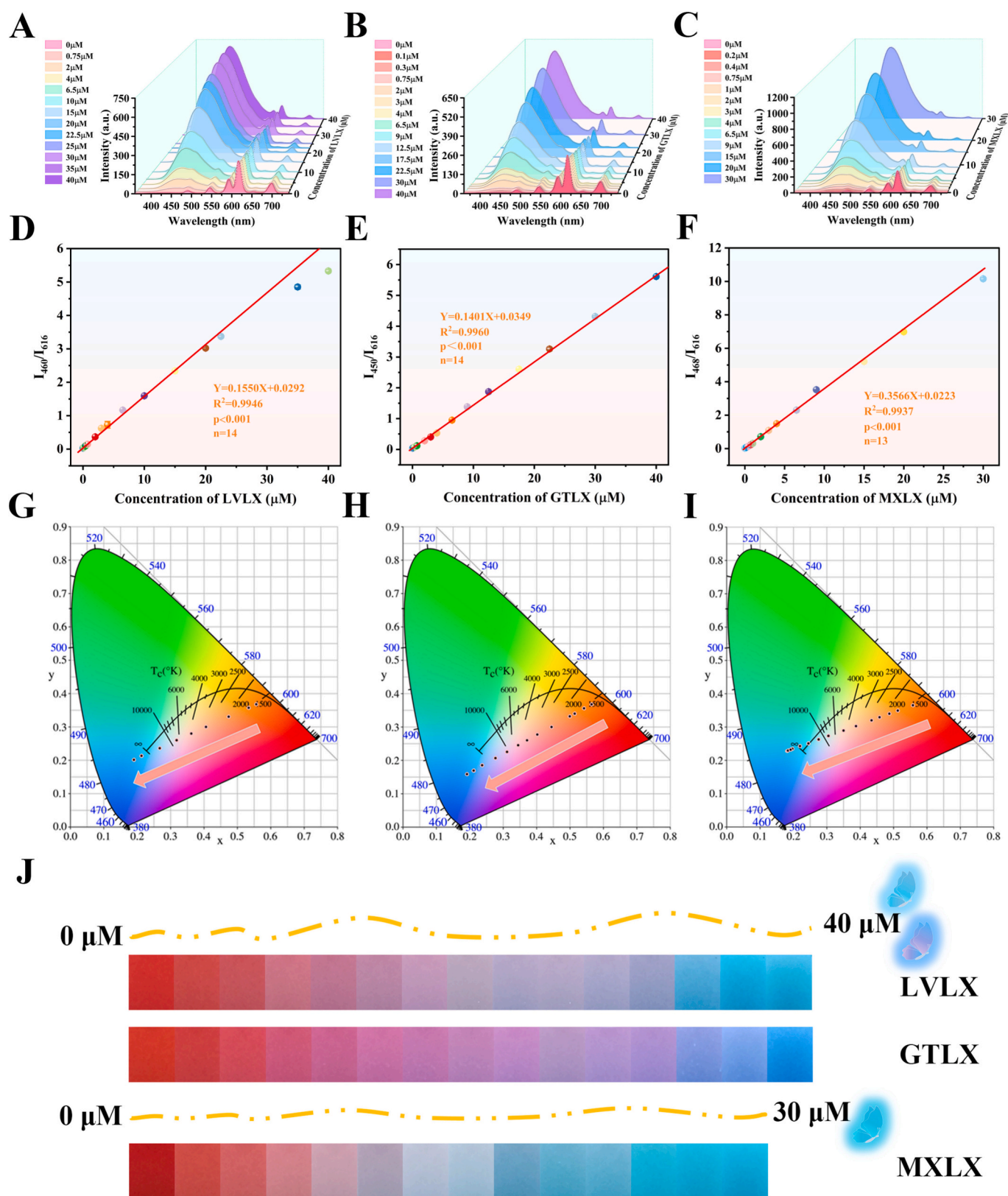


Fig. 3. The fluorescence spectra of GDP-Eu-Tb after adding different concentrations of LVLX (A), GTLX (B), and MXLX (C); the linear relationship between the fluorescence intensity ratio and the concentration of LVLX (D), GTLX (E), and MXLX (F); CIE chromaticity diagram of the GDP-Eu-Tb fluorescence probe at different QNs concentrations (G-I); Fluorescence images of GDP-Eu-Tb for LVLX, GTLX, and MXLX under a 302 nm ultraviolet lamp (J).

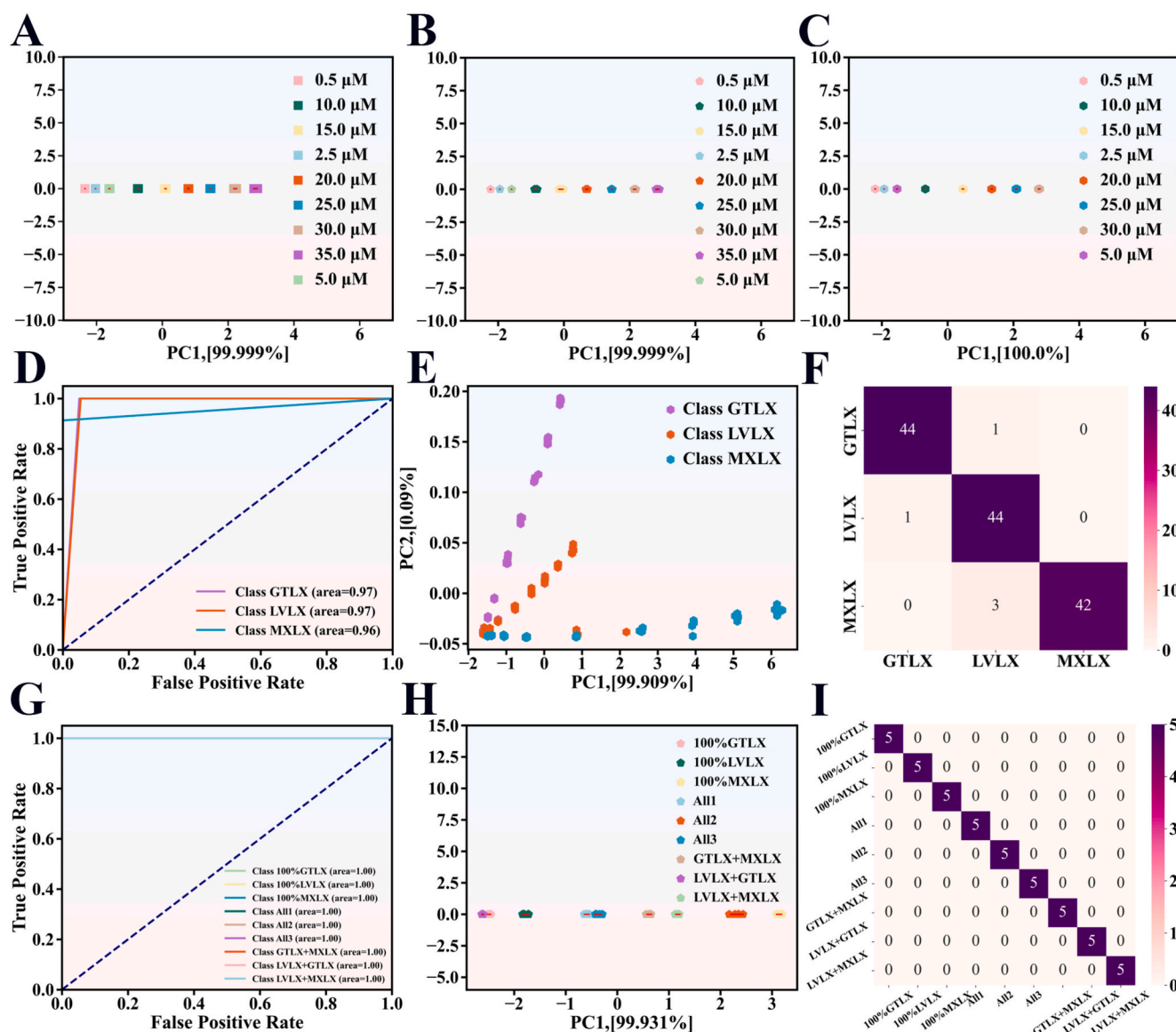


Fig. 4. Classification scatter plots predicted by the decision tree model based on unknown datasets with different concentrations of (A) LVLX, (B) GTLX, and (C) MXLX; (D) ROC curve, (E) classification scatter plot, and (F) confusion matrix generated based on unknown datasets with three different concentrations of QNs; (G) ROC curve, (H) classification scatter plot, and (I) confusion matrix generated based on different combinations of LVLX, GTLX, and MXLX concentrations (All1: 20 % LVLX + 60 % GTLX + 20 % MXLX; All2: 20 % LVLX + 20 % GTLX + 60 % MXLX; All3: 60 % LVLX + 20 % GTLX + 20 % MXLX).

distances between different prediction clusters are significantly larger. This indicates that the DT model not only accurately distinguishes between different antibiotic concentrations, but also fully exploits the main features extracted after PCA dimensionality reduction, ensuring stability and reliability in the predictions.

Furthermore, an analysis of the confusion matrix results for unknown samples (Figs. S11A-C) indicate that all non-zero values correspond with the main diagonal of the matrix, thus confirming a one-to-one correspondence between predicted and actual labels and underscoring the robust classification capability demonstrated by our model. Subsequently, the hierarchical clustering analysis (HCA) dendrogram (Figs. S11D-F), based on average-linkage methodology, is employed to illustrate the decision-making process regarding antibiotics across various concentrations. The HCA clusters each concentration level accurately. These findings demonstrate that the sensor array provides high-quality data, enabling the DT model to effectively distinguish between individual antibiotics at different concentration levels. The confidence interval is delineated through the third-party machine learning

library scikit-learn (sklearn) using Gaussian Mixture Models. One-dimensional data is visualized as line segments, while two-dimensional data is represented as ellipses. The characterization of confidence intervals intuitively reflects the distributional characteristics inherent to different data types.

To demonstrate the feasibility of fluorescence sensor arrays for analyzing QNs, we examine the identification of various antibiotic types, specifically LVLX, GTLX, and MXLX, across different concentration levels. The DT model evaluates LVLX, GTLX, and MXLX samples within a concentration range of 0.5 to 30 μM . As shown in Table S6 and Figs. S12A-B, the model using PC1 and PC2 (representing the first two principal components with significant cumulative contribution rates) demonstrates the best learning performance. Thus, these PCs are employed for model construction. The dimensionality reduction ensures that critical information is preserved while minimizing noise, which improves the robustness of the model. As shown in Fig. S13, the samples are classified by evaluating the value of a certain feature in the feature list, with the division constrained by entropy. A smaller entropy value

indicates a lower classification uncertainty. The entropy value corresponding to the leaf node is 0, signifying the end of the sample division process and indicating complete classification certainty. Fig. 4D indicates that the ROC curve trend approaches the upper left corner, and the area under the curve (AUC) value, presented in the lower right corner, is nearly equal to 1. This suggests that the DT model demonstrated excellent classification capabilities for differentiating varying concentrations of QNs (Lin et al., 2024). This model is subsequently applied to analyze unknown sets of samples at various concentrations. As depicted in Fig. 4E, the expected outcomes for different concentrations of LVLX, GTLX, and MXLX are dispersed along three distinct axes, confirming the model ability to distinguish structurally similar analytes. While there is some overlap between the points located in the lower left quadrant of Fig. 4E, several points deviate from their surrounding colors, indicating misclassification. Upon detailed analysis, Fig. 4F reveals that a data point intended for classification as GTLX was erroneously identified as LVLX. Similarly, one LVLX data point is incorrectly labeled as GTLX, while three MXLX data points are incorrectly predicted as LVLX. These errors most likely stem from inherent similarities in fluorescence properties among the analyzers, suggesting opportunities for additional optimization. Overall assessment shows that most non-zero values aligned closely with the main diagonal of the confusion matrix (Fig. 4F), with only a minimal number distributed on either side, which suggests an enhanced capability of the DT model to accurately predict the corresponding category. Through precise analysis across varying QN concentrations, it is evident that the DT-model-assisted ratiometric fluorescence sensor array can quantitatively distinguish structurally similar target analytes (Pushina et al., 2021).

Furthermore, the components present in actual environmental systems are highly complex. Scenarios in which only a single antibiotic is involved are rare, and the majority of instances involve the coexistence of multiple antibiotics. Given that fluorescence probes typically exhibit consistent responses (either synergistic or antagonistic) to such mixtures (Liu et al., 2024), the developed sensor array is employed to detect antibiotic combinations. Initially, based on the evaluation results from

the DT model constructed using varying numbers of PCs under stratified 10-fold cross-validation (Table S7 and Figs. S14A-B), PC1 and PC2 are identified as the most significant features for constructing prediction models, as they retained the majority of the variance in the data. These PCs effectively capture the distinct fluorescence response patterns of antibiotic mixtures, enabling precise differentiation of their concentration ratios. As illustrated in Figs. 5A-C, antibiotic mixtures with different concentration ratios are accurately predicted and distinguished. Notably, the confidence ellipses corresponding to antibiotics with varying mixing ratios do not overlap. This outcome reflects the effectiveness of the selected features and the accuracy of the model performance. The categories predicted by the DT model correspond one-to-one with their actual labels (Figs. 5D-F), further confirms the reliability of the approach. These prediction results indicate that DT models using 2D data pre-processed via PCA exhibit robust predictive performance for detecting two QN mixtures. Subsequently, based on further evaluation results (Table S8 and Figs. S15A-B), PC1 is chosen to construct a prediction model aimed at analyzing LVLX, GTLX, and MXLX across different concentration ratios while testing various QN mixtures. Utilizing this DT model yields an area under the curve (AUC) value of 1 for all tested combinations (Fig. 4G), signifying near-perfect classification capability. This reflects the robustness of the model in dealing with complex datasets, even when dealing with targets with similar structures. For assessing unknown samples comprising different combinations of three QNs, predictions made by the model accurately identify their respective categories (Fig. 4H), aligning perfectly with actual values observed (Fig. 4I). Collectively, these experimental findings demonstrate that arrays of fluorescence sensors, assisted by the DT model constructed in this study, have significant potential for simultaneous identification and discrimination of multiple QNs.

3.6. Anti-interference ability evaluation of GDP-Eu-Tb

The presence of other impurities in the complex, real-world detection environment may significantly affect the detection performance of the

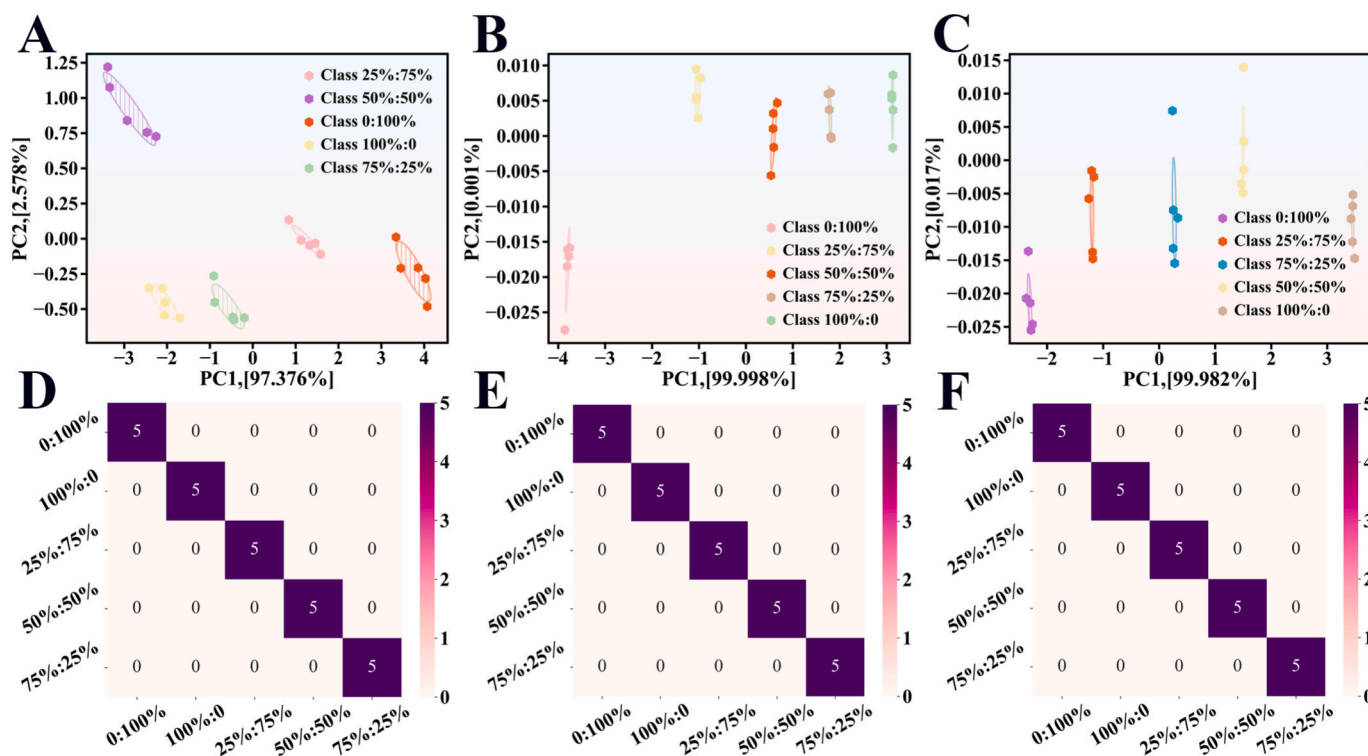


Fig. 5. The classification scatter plots (A–C) and confusion matrix (D–F) for unknown samples of LVLX and GTLX, LVLX and MXLX, and GTLX and MXLX with different concentration ratios.

material. To evaluate the effectiveness of the GDP-Eu-Tb fluorescence probe under actual conditions and to determine its specific anti-interference capability, we investigate its detection performance for three target QNs in the presence of common cations (Ca^{2+} , Pb^{2+} , Cd^{2+} , K^{+} , Ni^{2+} , Mg^{2+} , Cu^{2+} , Na^{+} , Fe^{3+} , and Ag^{+}) as well as anions (NO_3^- , Cl^- , SO_4^{2-} , and CH_3COO^-). As illustrated in Figs. 6A-C, GDP-Eu-Tb does not exhibit a significant fluorescence response to metal ions or anions before introducing QNs into the system. Upon introducing the target QNs into

this matrix, a slight fluctuation in the fluorescence intensity ratio is observed compared to the system without ion interference. This suggests that the common ions do not interfere with the fluorescence signal of the nanoprobe in the non-specific detection of LVLX, GTLX and MXLX, indicating that such fluorescence probes have remarkable anti-interference capabilities. This finding establishes a solid basis for employing it as a singular sensing element within an array of fluorescence sensors for the analysis and detection of LVLX, GTLX, and MXLX in

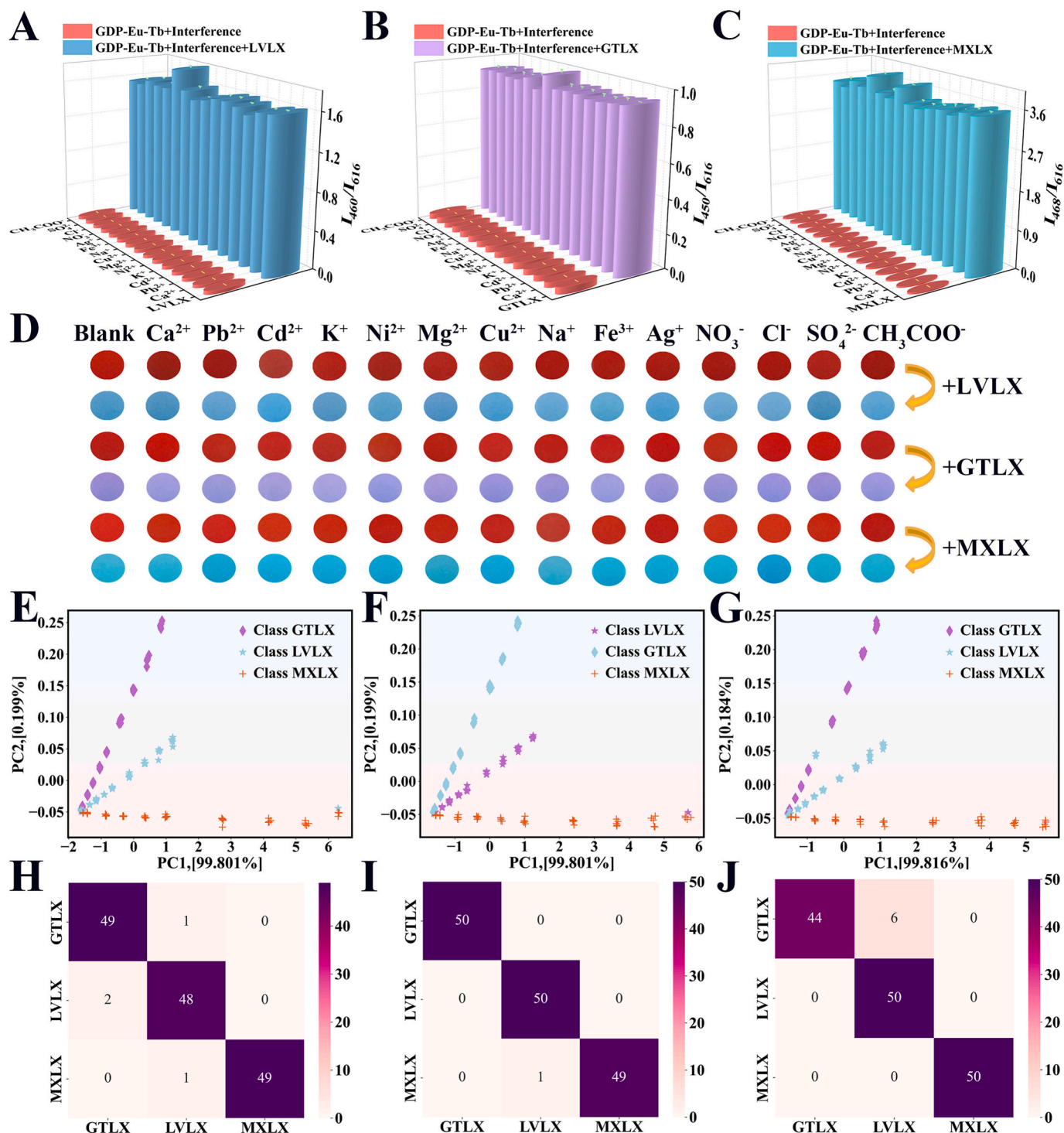


Fig. 6. The fluorescence intensity ratio of the GDP-Eu-Tb probe before and after adding LVLX (A), GTLX (B), and MXLX (C) and the corresponding fluorescence images under a 302 nm ultraviolet lamp (D); Decision tree modeling of QNs in cationic solutions (Ca^{2+} , Pb^{2+} , Cd^{2+} , K^{+} , Ni^{2+} , Mg^{2+} , Cu^{2+} , Na^{+} , Fe^{3+} , and Ag^{+}) (E), anionic solutions (NO_3^- , Cl^- , SO_4^{2-} , and CH_3COO^-) (F), and the actual honey sample solution (G) for the unknown samples (0.5 to 30 μM) with categorized scatter plots and corresponding confusion matrices (H)-(J).

complex environments. Fig. 6D shows fluorescence images captured before and after adding equal concentrations of LVLX, GTLX, and MXLX under irradiation with a 302 nm UV lamp in the presence of the interfering material. The results unambiguously demonstrate that the IFE is activated only upon interaction with the QNs, resulting in a quenching of the Eu^{3+} red fluorescence, which leads to a significant change in the fluorescence color visible to the naked eye.

To further evaluate the interference resistance of the fluorescence probe, we selected Fe^{3+} and Cu^{2+} , two common metal ions that may affect the probe's performance in environmental settings (Awual, Hasan, et al., 2019; Awual, Islam, et al., 2019). Furthermore, the impact of different honey concentrations on the probe's performance in the presence of interfering substances is examined. The changes in the fluorescence intensity ratio (I_{463}/I_{616}) under the interference conditions are calculated by comparing them to the blank system (Fig. S16). The results show that, in the metal ion interference environment (Fig. S16A), the fluorescence ratios change within a $\pm 5\%$ range, while in the honey environment (Fig. S16B), the change is about 13 %. Although the interference from honey is more significant, the changes remain within an acceptable range, indicating that the probe is suitable for detecting target substances in complex environments (Hidayat et al., 2017).

Effective anti-interference capability is crucial for the sensor array to obtain high-quality data, which is essential for constructing predictive models with robust generalization ability. Different concentrations of the three target QNs are introduced into cationic and anionic interference solvents at 10 μM , generating a dataset reflects an ionic interference environment. As shown in Table 1, accuracy, precision, recall, and F1 score (calculated using Eqs. (1)–(4)) are used to evaluate and compare the performance of the DT model under identical parameters across various datasets. The four evaluation metrics for the DT model on the test set marginally exceed those on the unknown dataset, with differences of less than 4 %. Moreover, the evaluation metric for unknown data remains around 0.9600, indicating that the model has commendable generalization ability. For the unknown samples obtained in this ion interference context, all four classification metrics of the DT model maintain values no lower than 0.9733. This suggests that the DT model based on this sensor array not only demonstrates strong predictive power, but also maintains good balance in the recognition of each category throughout its classification prediction process. As illustrated in Figs. 6E–F and 6H–I, these antibiotics can still be accurately identified within the complex matrix solution, highlighting that the developed array exhibits strong resistance to ionic interference.

3.7. Detection of QNs in honey

The potential of the GDP-Eu-Tb fluorescence probe in real-world environments is evaluated by performing recovery and detection experiments on QNs in honey using the standard addition method, following the treatment protocol outlined in Section 2.7. As shown in Table S9, the fluorescence probes exhibit a recovery rate ranging from 96.990 % to 118.696 % for QNs in actual honey samples, with a relative standard deviation (RSD) below 3.351 %. A linear regression analysis of

Table 1

Performance comparison of the same parameter decision tree model based on the test set and unknown detection samples.

Model parameter	Dataset	Accuracy	Precision	Recall	F1 score
criterion = 'entropy' max_depth = 15	Test data	1.0000	1.0000	1.0000	1.0000
	Unknown cation data	0.9733	0.9736	0.9733	0.9734
	Unknown anion data	0.9933	0.9935	0.9933	0.9933
	Unknown honey data	0.9600	0.9643	0.9600	0.9599

the spiked and detected concentrations indicate that the GDP-Eu-Tb fluorescence probe demonstrated reliable accuracy in detection LVLX and MXLX (Fig. S17A and Fig. S17C). For GTLX (Fig. S17B), the 95 % confidence interval of the regression slope is [1.0811, 1.1212], suggesting that the detected results might be slightly higher than the actual values within the concentration range of 0.5 to 25 μM . Such deviations may be due to experimental conditions or interactions between the probe and the biomarker. Nonetheless, a good linear relationship between the detected and spike concentrations is maintained. Therefore, the comprehensive evaluation demonstrates that the developed GDP-Eu-Tb fluorescence nanoprobe demonstrates high accuracy and reliability in detecting QNs in actual samples, indicating broad potential for practical applications.

Furthermore, we investigated the detection performance of a DT-assisted array of ratiometric fluorescence sensors on real samples. The DT model with the same parameters still exhibits robust predictive performance for unknown data in the honey environment (Table 1). This indicates that the model is stable and can consistently and efficiently predict based on the sensor array data across different interference environments. As shown in Fig. 6G and J, LVLX, GTLX, and MXLX with various levels are successfully distinguished. Such arrays demonstrate strong anti-interference capabilities and holds significant application potential. It is worth noting that testing environments are often complex. Unknown substances, ambiguous antibiotic types, and concentrations can disrupt optimal laboratory conditions and experimental designs. In order to achieve high-throughput identification and differentiation of antibiotics with minimal prerequisites, it is anticipated that real-time identification and differentiation of QNs and other antibiotics can be achieved by exploring more representative features and building high-performance models based on a rich dataset through ensemble learning and DL.

4. Conclusions

In conclusion, this study has developed a ratiometric fluorescence sensor array based on a nucleotide-bimetallic lanthanide coordination polymer (GDP-Eu-Tb) as a single sensing element, aimed at achieving efficient detection of QNs. Under initial red light conditions, the sensor enables ratiometric fluorescence detection of QNs through the IFE, where QNs quench the fluorescence signal of Eu^{3+} and simultaneously enhance the intensity of their own characteristic peaks, thereby improving the fluorescence signal response. Compared to traditional single fluorescence probes, the ratiometric fluorescence sensor array developed in this study demonstrates significant advantages in detection accuracy and anti-interference capability, while also offering a simpler preparation and operation process. By integrating PCA and DT algorithms, this study successfully enables the simultaneous identification of LVLX, GTLX, and MXLX, and accurately distinguishes various QN combinations. Compared to existing methods based on single-signal fluorescence probes or traditional sensor arrays, this study constructs a ratiometric fluorescence sensor array using the GDP-Eu-Tb complex, combined with ML algorithms, to enhance the accuracy and stability of detection results. Particularly in various interference environments, such as QN detection in honey, the sensor array exhibits strong adaptability and robustness. The introduction of ML provides new approaches for signal processing and data analysis, contributing to the efficient recognition and classification of different types of biomarkers. Overall, this study offers an innovative solution for the rapid detection of QNs and demonstrates the broad application potential of the ratiometric fluorescence sensor array in complex environments.

CRediT authorship contribution statement

Mengyuan Li: Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Lei Jia:** Writing – review & editing, Writing – original draft, Supervision, Project administration,

Methodology, Funding acquisition. **Xiaolei Zhao:** Visualization, Validation, Software, Methodology. **Lina Zhang:** Visualization, Software, Formal analysis, Data curation. **Dan Zhao:** Validation, Software, Methodology, Conceptualization. **Jun Xu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Tongqian Zhao:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2025.143722>.

Data availability

No data was used for the research described in the article.

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